COMP301 (2026): Software Design

Confident Agent — Project Proposal

An Adaptive Tutoring Agent with Confidence-Aware Content Validation

1. Introduction

Adaptive tutoring systems — agents that analyse a student's quiz performance, identify weak areas, and generate a personalised learning path of lessons and practice quizzes — are a natural and valuable application of Large Language Models (LLMs) in education. However, this application is also highly exposed to LLM hallucination: an LLM asked to explain a concept, write quiz questions, or recommend reading material can produce content that is fluent and plausible but factually wrong, inconsistently graded, or entirely fabricated. In an educational setting, an unnoticed hallucination is particularly damaging, since the student is, by definition, not yet in a position to recognise the error.

This project investigates whether an agent-based validation architecture — a "Confident Agent" — can detect and reduce hallucinations in an LLM-driven adaptive tutoring pipeline, compared against a single-prompt baseline tutor with no validation.

2. Research Question

Can an agent-based validation architecture detect or prevent unacceptable hallucinations in AI-generated educational content (lessons, quiz questions and answer-keys, and recommended learning materials) before they reach the student, compared with a baseline LLM tutor operating without validation?

3. Scientific Hypothesis

An agent-based validation architecture equipped with explicit confidence estimation (via retrieval-based verification and/or multi-agent consensus) will significantly reduce the rate of unacceptable hallucinations in generated lesson content, quiz answer-keys, and recommended learning materials, compared with an equivalent baseline LLM tutor operating without validation.

4. Application Domain

Domain: Educational tutoring systems — specifically, adaptive learning-path and quiz generation for a bounded subject area (final subject scope, e.g. a first-year Computer Science or Mathematics topic set, to be confirmed once benchmark construction begins). This domain is listed as a suggested application domain in the course brief and allows for objective ground-truth checking of both factual content and quiz correctness.

5. System Concept

The system takes a student's quiz history and performance data as input, identifies the topics the student is struggling with, and generates:

- a personalised learning path across the weak topics;
- lesson content explaining each topic;
- practice quiz questions (with answer-keys) targeting each topic;
- recommended learning materials (readings, videos, references) for further study.

The core research contribution is not the tutoring interface itself, but a comparison between generating this content with an unvalidated baseline LLM versus generating it through a Confident Agent pipeline that estimates confidence and validates content before it is shown to the student.

6. Hallucination Types Under Investigation

We focus on three hallucination types, chosen because each permits objective, checkable evaluation:

Hallucination Type	Where It Occurs	How We Detect It
Factual errors in lesson content	Generator Agent explains a concept incorrectly	Verifier Agent cross-checks claims against a curated reference source / consensus across independent generations
Incorrect quiz answer-keys	Generator Agent marks the wrong option as correct, or writes an ambiguous distractor	Verifier Agent independently re-solves the question and checks agreement
Fabricated learning materials	Suggestion Agent recommends a book, article or link that does not exist or misattributes a source	Verifier Agent checks existence via a search tool / bibliographic API (e.g. Open Library, Google Books, arXiv) before surfacing it

7. Agent Architecture

The system follows a Chief Agent coordination pattern, consistent with the architecture suggested in the course brief, separating content generation from independent validation:

Agent	Responsibility
Chief Agent	Coordinates the pipeline, interacts with the student, and decides what content is ultimately surfaced (accept, regenerate, or flag as low-confidence).
Generator Agent	Produces the candidate lesson explanation, quiz questions/answer-keys, and a personalised learning path from the student's performance data.
Suggestion Agent	Recommends learning materials to address the student's weak areas — either matched from a curated, pre-vetted resource pool, or generated openly and passed to the Verifier.
Verifier Agent	Independently checks factual claims, quiz answer-keys, and (where applicable) the existence of suggested materials, via retrieval, cross-model consensus, or tool-based lookup.
Confidence Estimator	Produces a numeric confidence score per content unit (lesson chunk, quiz question, or suggested resource) based on verifier agreement / consensus voting.

Materials recommendation will be evaluated under two conditions, giving us an additional experimental axis: (A) a constrained Suggestion Agent that only recommends from a curated, pre-vetted resource pool (a retrieval/ranking problem), and (B) an open-generation Suggestion Agent whose recommendations are checked post-hoc by the Verifier Agent via a search/bibliographic API. Comparing these two conditions lets us report on whether grounding at generation time outperforms post-hoc verification, or vice versa.

Note on infrastructure: we do not currently plan to use a vector database. Since the reference material for verification will be a bounded, curated set of documents per topic, we will use direct topic-keyed lookup rather than embedding-based retrieval. A vector database (and full RAG) is a possible extension if the benchmark or reference corpus grows large enough to require semantic search; the choice of technology will not be assessed directly, so this is a scope decision rather than a limitation.

8. Experimental Methodology

8.1 Benchmark Construction

We will construct a benchmark of approximately 50–100 prompts spanning the chosen topic set, varying in difficulty and topic obscurity (obscurity is expected to correlate with hallucination rate, giving us a variable to analyse). For each prompt we will establish ground truth (via textbook/curriculum reference or manual verification) so that lesson-content correctness, quiz answer-key correctness, and material-recommendation existence can all be scored objectively rather than subjectively.

8.2 Baseline Evaluation

The benchmark will first be run against a single-prompt baseline LLM tutor with no validation pipeline, and errors will be recorded per hallucination type described above.

8.3 Comparative Evaluation

The identical, unmodified benchmark will then be run through the completed Confident Agent. We will measure not only the reduction in hallucination rate, but also unintended side effects — specifically, the false-rejection rate (the Confident Agent incorrectly flagging correct content as untrustworthy), since over-cautious validation is itself a usability cost.

8.4 Statistical Analysis

Since each benchmark prompt produces a paired binary outcome (correct/incorrect) under both the baseline and the intervention, we plan to use a paired test appropriate to this design (e.g. McNemar's test) to determine whether the observed reduction in hallucinations is statistically significant, alongside confidence intervals rather than point estimates alone.

8.5 Threats to Validity

We anticipate discussing the following limitations in the final report:

- benchmark size and topic scope may limit generalisability to other subjects;
- ground-truth scoring performed by the group introduces potential evaluator bias;
- the Verifier Agent's own reference source may itself be incomplete, which could understate or overstate hallucination rates;
- model drift — results are tied to the specific LLM versions used at the time of evaluation;
- practical trade-offs introduced by validation, including additional latency, API cost, and implementation complexity, will be reported alongside accuracy gains.

9. Project Timeline

Milestone	Expected Outcome / Our Focus
Project Proposal	Application domain selected (EdTech tutoring); research question and hallucination types defined (this document).
Benchmark Design	Hallucination benchmark constructed: lesson-content factual errors, quiz answer-key errors, fabricated learning materials.
Baseline Evaluation	Baseline (single-prompt) LLM tutor evaluated against the benchmark; error rates recorded.
System Implementation	Confident Agent pipeline implemented: Generator, Verifier, Confidence Estimator, Suggestion Agent, Chief Agent.
Experimental Evaluation	Identical benchmark run through the Confident Agent; baseline vs. intervention compared.
Final Submission	Report, source code, benchmark and presentation submitted.

10. What We Aim to Learn

Beyond the research result itself, this project is intended to give us hands-on experience with:

- designing and coordinating a multi-agent architecture (Chief Agent + specialised sub-agents);
- tool calling and external validation (bibliographic/search APIs for reference checking);
- confidence estimation strategies (cross-model consensus and/or retrieval-based verification);
- constructing a domain-specific evaluation benchmark with objective scoring;
- applying statistical analysis to evaluate an AI system, rather than relying on informal impressions of quality.